MD ANISUR RAHMAN CHOWDHURY

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PROFESSIONAL SUMMARY

Network security and infrastructure professional with 7+ years of experience spanning ISP operations, enterprise IT, systems administration, and production support. Hands-on background in firewall management, routing and switching, VPN design, Linux/Windows Server, and Active Directory. Extended expertise into machine learning-based threat detection, serverless security architecture, LLM development, and AI workflow automation through active research and published IEEE work. M.S. Information Technology candidate at Gannon University (GPA: 3.93). CCNA, JNCIA, and MTCNA certifications.

TECHNICAL SKILLS

Networking: TCP/IP, VLANs, BGP, OSPF, NAT, ACLs, VPN (IPsec/L2TP/PPTP), Load Balancing, WAN Failover

Security: Firewall policy management, Zero Trust design, IDS/IPS, IAM, WAF, threat detection, incident response

Systems: Windows Server, AD DS, Linux (Ubuntu/RHEL/CentOS/AlmaLinux), DNS, DHCP, Postfix/Dovecot

Cloud & Virt: AWS (Lambda, IAM, WAF, EC2, S3), Proxmox, VMware ESXi, Hyper-V

Devices: Cisco, MikroTik, Juniper, Fortinet, Cyberoam

ML & LLM: LSTM, BiGRU, XGBoost, CNN, SVM; model training/evaluation; self-correcting LLM frameworks; multi-agent ensemble orchestration; Ollama local inference

AI Automation: n8n workflow design and self-hosted deployment; dual-engine pipelines (Gemini + Ollama); async job processing; webhook integrations

Prompt Eng.: System prompt construction, chain-of-thought structuring, structured JSON output design, iterative model refinement

Dev Tools: Python, Java, SQL/PLSQL, JavaScript, Next.js, React, Tailwind CSS, TypeScript, Chart.js, Firebase Authentication + Firestore, Cloudflare, Github;

CERTIFICATIONS

CCNA – 2024 | Cisco Certified Network Associate **JNCIA – 2022** | Juniper Networks Certified Associate **MTCNA – 2019** | MikroTik Certified Network Associate

PROFESSIONAL EXPERIENCE

Graduate Research Assistant | Gannon University – CIS Dept., Erie, PA *Aug 2025 – May 2026*

- Developed prototype logic for secure data transmission optimization between data centers; designed secure path reservation algorithms and real-time traffic flow models for enterprise-scale network environments.
- Trained and evaluated LSTM, BiGRU, and hybrid XGBoost intrusion detection models against the CICIDS2017 benchmark; co-authored conference publications on network security, cloud infrastructure, and LLM reliability.

Cybersecurity Lecturer (CPT) | TAAS Services LLC, Remote, USA *Aug 2025 – Dec 2025*

- Delivered 15+ hands-on cybersecurity labs covering network defense, threat detection, firewall configuration, and vulnerability management; developed certification-aligned curriculum and practical lab exercises.

Senior Executive – Information Technology | S. Alam Group, Chittagong, Bangladesh *Dec 2021 – Jul 2024*

- Administered hybrid Linux/Windows server environments supporting 500+ enterprise users — mail services (Postfix/Dovecot), web applications, and critical business systems; managed AD DS for user provisioning, GPO enforcement, and RBAC across multiple departments.
- Maintained IP-PBX/VoIP systems, planned and executed LAN/WAN infrastructure upgrades, and monitored network performance using LibreNMS — reducing call-quality incidents and unplanned downtime while maintaining SLA-level uptime.

Technical Support Engineer – Networking | Link3 Technology Ltd., Chittagong, Bangladesh *Aug 2017 – Dec 2021*

- Provided Tier 2/3 support across Cisco, MikroTik, Juniper, Fortinet, and Cyberoam devices for 100+ enterprise accounts — configuring firewall policies, NAT, ACLs, IPsec/L2TP VPN tunnels, load balancing, and WAN failover.
- Troubleshot radio-link and fiber faults, implemented structured escalation workflows that measurably reduced MTTR, and documented network topologies, configuration standards, and incident runbooks.

EDUCATION

M.S., Information Technology
Gannon University, Erie, PA | GPA: 3.93/4.00 | *Expected May 2026*

B.S., Electronics & Telecommunication Engineering
University of Science & Technology Chittagong, Bangladesh | 2017

AI, MACHINE LEARNING & AUTOMATION

- Designed and refined system prompts, chain-of-thought templates, and structured JSON output schemas for production LLM deployments; developed the three-stage Generator–Critic–Synthesizer prompt framework for the Self-Correcting LLM project, achieving a 60% relative accuracy improvement on the GSM8K benchmark (31.2% → 49.9%) with a 78% error correction rate.
- Trained a 3-layer LSTM model on CICIDS2017 for multi-class network intrusion detection — 98% accuracy, precision, recall, and F1 across five attack categories (IEEE CSCloud 2025); extended into a hybrid XGBoost + BiGRU cross-cloud architecture with 0.990 ROC-AUC and 135 ms avg. response latency (IEEE SmartCloud 2026, under review).
- Deployed a distributed multi-agent LLM ensemble using four local Ollama-based models (Llama3.2, Qwen2.5, Phi3, Gemma2) coordinated by a FastAPI orchestrator; implemented majority voting, dynamic weighted voting, ISP routing, and two-round debate strategies; benchmarked across MMLU, GSM8K, and TruthfulQA with paired t-tests and Wilcoxon signed-rank tests.

- Designed and deployed six production n8n automation workflows supporting a dual-engine architecture (Google Gemini + self-hosted Ollama): food image analysis with calorie/macro extraction, 7-day personalized meal planning, and weekly workout generation — using UUID-based async job processing and webhook-triggered pipelines.
- Built and deployed DataMentor, a serverless CSV intelligence platform with browser-native Python execution (Pyodide), deterministic cleaning pipelines, automated notebook generation, Firebase auth, and a hybrid rule/model AI repair layer for runtime fault recovery.

RESEARCH PUBLICATIONS

Towards a Serverless Intelligent Firewall: AI-Driven Security and Zero-Trust Architectures
IEEE CSCloud 2025 — Lead author; LSTM-based intrusion detection; 98% accuracy on CICIDS2017; serverless Zero-Trust on AWS Lambda.

Towards a Serverless Intelligent Firewall: Cross-Cloud Adaptation, AI-Driven Security, and Zero-Trust Architectures
IEEE SmartCloud 2026 (Under Review) — Lead author; hybrid XGBoost + BiGRU engine; multi-cloud Zero-Trust policy orchestration.

Teaching Large Language Models to Think Twice: A Three-Stage Self-Correction Framework
Conference Submission — Lead author; Generator–Critic–Synthesizer framework; 60% relative accuracy gain on GSM8K; 78% error correction rate.

A Local Distributed Multi-Agent LLM Ensemble System
Conference Submission — Lead author; FastAPI orchestrator + four Ollama agents; five aggregation strategies; MMLU/GSM8K/TruthfulQA benchmarks.

Auction-Based Dynamic Resource Allocation for Optimized Edge Computing in Distributed Networks
IEEE CSITSS 2025 — Proposed auction-theoretic model for efficient resource allocation in distributed edge computing.

AI and Cloud Computing in Business Systems: A Hybrid Model for Enhancing Enterprise Resource Planning
IEEE CSITSS 2025 — Designed and evaluated an AI-driven cloud ERP framework integrating machine learning to enhance enterprise decision-making.

KEY PROJECTS

Serverless Intelligent Firewall – Research Series (AWS) [live]
- Event-driven cloud security using AWS Lambda, IAM, and WAF with LSTM-based and hybrid BiGRU/XGBoost detection; automated threat response workflows. GitHub: Research-1 / Research-2 / Research-3.

Self-Correcting LLM – Three-Stage Reasoning Framework [live]
- Generator–Critic–Synthesizer framework enabling local LLMs to detect and correct multi-step reasoning errors; 60% relative accuracy improvement on GSM8K; 78% error correction rate.

Distributed Multi-Agent LLM Ensemble [live]
- FastAPI orchestrator routing prompts to four local Ollama agents with five aggregation strategies; full statistical benchmarking across MMLU, GSM8K, and TruthfulQA.

DataMentor – Serverless CSV Intelligence Platform [live]
- Browser-native analytics with Pyodide Python execution, deterministic CSV cleaning, automated notebook generation, Firebase auth, and a hybrid AI repair layer for runtime fault recovery.

FitLife + n8n AI Automation Engine [live]
- Six production n8n workflows delivering food image nutritional analysis, 7-day meal planning, and weekly workout generation via dual-engine AI (Google Gemini + self-hosted Ollama).

Secure Mail Server & Active-Active DC/DR Lab
- Production-grade Postfix/Dovecot/Roundcube stack with full DNS hardening (SPF, DKIM, DMARC); high-availability Proxmox virtualization lab with DR planning, failover testing, and real-time monitoring.

LANGUAGES

English (Professional Proficiency) | Bengali / Bangla (Native)